DATA CLEANING REPORT

HR Raw Dataset – Power Query Transformation

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1. Introduction & Objective

This report documents the complete data cleaning and feature-engineering process applied to the raw Human Resources (HR) employee dataset using Microsoft Power Query (Power BI / Excel). The goal was to convert a messy, inconsistent raw file into a clean, analysis-ready dataset and then enrich it with derived columns that support dashboards, attrition analysis, salary benchmarking, tenure studies, and performance insights.

All transformations were performed in the Power Query Editor to ensure reproducibility, transparency, and an auditable pipeline.

2. Column Overview

Original Columns: Employee_ID / Full_Name / Department / Job_Title / Hire_Date / Performance_Rating / Experience_Years / Status / Work_Mode / Salary / Year / Country / City / Age / Job_Level

Derived Columns Added: / Hire_Year ,Tenure_Years, Years_Since_Hire, Is_Active, Performance_Score, Salary_Band, Hire_Month, Hire_Quarter

3. Data Quality Issues Identified

A thorough profile of the raw data revealed multiple quality problems that would have compromised any analytical work if left unaddressed:

#	Issue Category	Description & Impact
1	**Inconsistent Casing**	Names, departments, job titles, status, and ratings appeared in mixed case (e.g., "dylan wilkinson", "XAVIER WILSON", "operations", "excellent").
2	**Department Name Variants**	Multiple spellings and abbreviations: "Ops", "Sale", "Hr", "IT", "Human Resources", "operations" – preventing proper grouping.
3	**Mixed Date Formats**	Hire_Date contained DD-MM-YYYY, MM/DD/YYYY, YYYY/MM/DD, and text formats (e.g., "08 Mar 2022"), making date calculations impossible.
4	**Currency Symbols in Salary**	Salary values included "€" and "EUR" text (e.g., "53007 EUR"), preventing numeric aggregation and statistical analysis.
5	**Status Inconsistencies**	Values such as "active", "ACTIVE", "Currently", "RESIGNE", "Terminated" with varying capitalization and truncation.
6	**Work Mode Variants**	"Onsite", "On-site", "On-Site", "Remote", "Hybrid" – inconsistent spelling and casing.
7	**Performance Rating Noise**	Mixed case and prefixed values (e.g., "3 - Satisfactory", "excellent", "Needs Improvement").
8	**Leading/Trailing Spaces**	Extra whitespace in text columns caused lookup failures and incorrect distinct counts.
9	**Invalid Year Values**	Year column contained outliers such as 25, 1999, 2027 – inconsistent with Hire_Year.

4. Data Cleaning Steps (Power Query)

All cleaning transformations were applied as sequential steps in the Applied Steps pane. The major steps are detailed below.

4.1 Trim Text Columns

Action: Transform → Format → Trim

Applied to Full_Name, Department, Job_Title, City and other text columns to remove leading/trailing whitespace that would otherwise create duplicate categories.

4.2 Capitalize Each Word (Proper Case)

Action: Transform → Format → Capitalize Each Word

Converted mixed-case strings into consistent Proper Case (e.g., "dylan wilkinson" → "Dylan Wilkinson", "XAVIER WILSON" → "Xavier Wilson").

4.3 Standardize Department Names

Action: Replace Values (multiple sequential steps). Key mappings:

Value To Find	Replace With	Notes
Sale	Sales	Abbreviation fix
Ops	Operations	Abbreviation expansion
Hr / Human Resources	HR	Standardized short form
IT / Information Technology	IT	Standardized short form
operations (lowercase)	Operations	Case + proper name

4.4 Clean Performance Rating

- "excellent" / "Excellent" → Excellent
- "satisfactory" → Satisfactory
- "3 - Satisfactory" → cleaned to proper rating
- Final values: Excellent, Good, Satisfactory, Needs Improvement

4.5 Standardize Status Column

- "active" / "ACTIVE" / "Currently" → Active
- "RESIGNE" and residual variants → Resigned
- Final values: Active, Resigned, Terminated

4.6 Fix Work Mode Values

- "Onsite" / "On-Site" → On-site
- Final values: On-site, Hybrid, Remote

4.7 Clean Salary Column (Remove Currency Symbols)

Action: Replace Values → Replace "€" and "EUR" with blank.

Example: "53007 EUR" → 53007. Salary became a pure numeric column ready for aggregation and banding.

4.8 Standardize Hire Date (Mixed Formats)

Action: Add Column → Custom Column (new_hire_date)

M Formula used:

```
try Date.FromText([Hire_Date], "en-GB") otherwise try Date.FromText([Hire_Date], "en-US") otherwise
null
```

This robust approach successfully parsed the majority of mixed European (DD-MM-YYYY) and US (MM/DD/YYYY) date formats into a true Date type.

5. Feature Engineering – Derived Columns

After the core cleaning steps, additional calculated columns were created to support deeper HR analytics. These derived fields enable tenure analysis, performance scoring, salary banding, and time-based hiring trend studies without requiring repeated calculations in every visual.

5.1 Job_Level

Purpose: Categorize employees into hierarchical levels for workforce structure and compensation analysis.

Values: Junior, Mid, Senior, Director

Logic: Assigned based on Job Title seniority patterns and existing data distribution (e.g., titles containing “Director” or “CFO” preferentially mapped to Senior/Director levels).

5.2 Hire_Year

Purpose: Extract the calendar year of hire for trend analysis and year-over-year comparisons.

Derivation: Year component extracted from the cleaned hire_date (Date.Year).

M example: Date.Year([new_hire_date])

5.3 Tenure_Years

Purpose: Measure length of service in years (with one decimal precision) as of the analysis reference date (7 August 2026).

Calculation: (Reference Date − hire_date) / 365.25, rounded to 1 decimal. Future hires set to 0.

Business use: Attrition risk, loyalty analysis, experience vs performance correlation.

Typical range observed: 0.0 – 18.6 years (mean ≈ 9.3 years).

5.4 Years_Since_Hire

Purpose: Simple integer tenure based on calendar years (2026 − Hire_Year), clamped at zero for future hires.

Complements Tenure_Years when only whole-year buckets are required in visuals or models.

5.5 Performance_Score

Purpose: Convert the categorical Performance_Rating into a numeric scale for averaging, ranking, and statistical modeling.

Performance Rating	Performance Score
Excellent	4
Good	3
Satisfactory	2

Needs Improvement	1

This numeric mapping enables average performance by department, correlation with salary/tenure, and easy filtering in dashboards.

5.6 Salary_Band

Purpose: Group continuous salary values into meaningful bands for distribution analysis and equity reviews.

Logic (approximate thresholds used):

- Low → salary ≤ 55,000
- Medium → 55,001 – 95,000
- High → salary > 95,000

Bands support compensation benchmarking, pay equity analysis, and simple visual segmentation without exposing exact salary figures.

5.7 Hire_Month & Hire_Quarter

Purpose: Enable seasonal and quarterly hiring pattern analysis.

Derivation:

- Hire_Month = Month number extracted from hire_date (1–12)
- Hire_Quarter = Quarter calculated from month ((Month−1) ÷ 3 + 1) → 1–4

Useful for identifying peak hiring seasons, quarterly headcount growth, and planning recruitment capacity.

6. Before vs After – Illustrative Examples

Column	Before (Raw)	After (Cleaned / Derived)
Full_Name	dylan wilkinson / XAVIER WILSON	Dylan Wilkinson / Xavier Wilson
Department	Ops / Sale / Hr / operations	Operations / Sales / HR
Hire_Date	08 Mar 2022 / 02/15/2019 / 18.04.2023	Proper Date type (unified)
Performance_Rating	excellent / 3 - Satisfactory	Excellent / Satisfactory + Score 4/2
Status	active / ACTIVE / RESIGNE	Active / Resigned / Terminated
Work_Mode	Onsite / On-Site	On-site / Hybrid / Remote
Salary	53007 EUR / values with €	53007 (numeric) + Salary_Band
Tenure / Scores	(not present)	Tenure_Years, Years_Since_Hire, Performance_Score, Hire_Month, Hire_Quarter

7. Summary of Applied Steps (Power Query)

The Applied Steps pane recorded the following major transformation sequence:

1. Source – Loaded CSV (UTF-8, comma-delimited)
2. Promoted Headers
3. Changed Type (initial type detection)
4. Removed Columns (unnecessary columns if any)
5. Trimmed Text – removed leading/trailing spaces
6. Capitalized Each Word – Proper case on name & categorical columns
7. Multiple Replace Value steps – Department, Status, Performance, Work Mode, Salary currency

8. Added Custom Column (new_hire_date) – robust date parsing
9. Added / refined Job_Level
10. Added Hire_Year (from cleaned date)
11. Added Tenure_Years
12. Added Years_Since_Hire (integer year difference)
13. Added Performance_Score (1–4 numeric mapping)
14. Added Salary_Band (Low / Medium / High)
15. Added Hire_Month and Hire_Quarter for time-series analysis

8. Conclusion

The Power Query process successfully transformed a highly inconsistent raw HR file into a reliable, structured, and analytics-ready dataset. Every major quality issue—casing, abbreviations, mixed dates, currency symbols, and free-text variations—was systematically addressed. In addition, a rich set of derived columns (Job_Level, Hire_Year, Tenure_Years, Years_Since_Hire, Performance_Score, Salary_Band, Hire_Month, Hire_Quarter) was engineered to accelerate downstream analysis. The resulting dataset is now suitable for advanced HR analytics including attrition modeling, compensation analysis, workforce planning, and performance dashboards.